



Week 4 Knowledge Check

Technical Architecture & Constraints

Technical Product Manager Academy



How This Works

12 questions across the week — architecture stance, STRIDE & compliance, the C4 model, SLOs & error budgets, and trade-offs.

Commit to an answer **before** the reveal. Retrieval is the point.

? Question 1

A multi-deploy system whose services *must* all be released together to work is best described as a:

- A) Modular monolith
- B) Service-oriented architecture (SOA)
- C) Distributed monolith
- D) Event-driven architecture



Answer 1 — C

Distributed monolith.

The anti-pattern: many deploys that still must ship together. You pay the ops cost of separation and gain none of the independence.

? Question 2 — True / False

In the architecture stance template, writing "**Revisit if:** later / when needed" is an acceptable revisit trigger.

Answer 2 — False

"Later" or "when needed" is **not a trigger**.

A real trigger is a concrete metric or org change — "if traffic grows 10x" or "if a second team starts contributing."

? Question 3

In a STRIDE pass, which letter is the *most-missed* — the least intuitive one that teams building without audit trails routinely skip?

- A) Spoofing
- B) Repudiation
- C) Information disclosure
- D) Elevation of privilege



Answer 3 — B

Repudiation. The least intuitive letter — denying you did something you did.

Teams that ship without audit trails miss it. A STRIDE list with zero R's should be pushed on.

? Question 4 — True / False

A strong security brief opens by telling the security team "we don't know anything yet, please tell us what to do."

Answer 4 — False

A passive "please tell us everything" brief signals an **outsourced-thinking TPM**.

Strong brief: short (5-min read), specific ("open questions" are concrete asks), and confident ("decisions we've made" with justification).

? Question 5

In the C4 model, which two levels are the TPM's scope to draw?

- A) Context and Container
- B) Container and Component
- C) Component and Code
- D) Context and Component



Answer 5 — A

Context and Container.

Component and Code belong to the architect and engineers. Staying above Component keeps the TPM in the right altitude.

? Question 6

Stress-testing a Container diagram with the three lenses (failure, trust boundary, evolvability), which annotation is the *most-missed* — where compliance and security risk concentrates?

- A) The failure path
- B) The trust boundary
- C) The evolvability arrows
- D) The async/sync distinction

Answer 6 — B

The trust boundary.

Where data crosses from our control to someone else's — mark it with a thick line. It's the most-missed annotation and where security and compliance risk concentrates.

? Question 7 — True / False

Under a modular-monolith stance, the modules should be drawn inside one container as labeled sub-boxes — not promoted to separate containers.



Answer 7 — True

A modular monolith is **one container**; draw the modules as sub-boxes inside it.

Microservices are many containers. The diagram should make the stance visible at a glance.

? Question 8

Which of these is a *contractual commitment with consequences* — as opposed to an internal target?

- A) SLI
- B) SLO
- C) SLA
- D) Error budget

Answer 8 — C

SLA — contractual, with consequences.

SLI = the measurement. SLO = the internal target. Error budget = $(1 - \text{SLO})$ over the window. Casually writing "we commit to 99.99%" turns an SLO into an SLA.

? Question 9

If a feature's availability SLO is 99.9% over a 30-day window, its error budget is roughly:

- A) 4 minutes
- B) 43 minutes
- C) 7 hours
- D) 22 hours



Answer 9 — B

~43 minutes.

Error budget = $(1 - 0.999) = 0.1\%$ of 30 days $\approx$ 43 minutes of allowed downtime. Inside the budget $\rightarrow$ ship; near/past it $\rightarrow$ prioritize reliability.

? Question 10 — True / False

Stating a latency SLO as "avg < 500ms" is a solid target because the average captures typical user experience.

Answer 10 — False

Average **hides bimodal distributions.**

Force percentiles (e.g., p95). Every SLO needs a percentile, a window, and a defense.

? Question 11

Which of these is a *mature* trade-off statement rather than a mere description?

- A) "We considered microservices but chose a monolith."
- B) "We chose a monolith because monoliths are simpler."
- C) "Option A: modular monolith. Option B: extract a service. We chose A. Accepted cost: future extraction needs a coordinated refactor. Revisit if traffic exceeds 5× rest-of-app."
- D) "We'll start with a monolith and figure out the rest later."

Answer 11 — C

Option A vs Option B → choice → accepted cost → revisit trigger.

"We chose X because X is better" is not a trade-off. A mature trade-off names what you gave up and when you'd change your mind.

? Question 12

When writing the top-5 architectural trade-offs, what is the recommended coverage check to confirm you surfaced real tensions?

- A) All 5 come from a single category
- B) They span at least 3 of the five trade-off categories
- C) Each names a specific competitor
- D) Each has a dollar figure attached

Answer 12 — B

Span at least 3 of the five categories.

Coupling vs independence · consistency vs availability · latency vs durability
· speed vs robustness · generality vs simplicity. All in one category usually means real tensions were missed.



That's Week 4

Score yourself — but the real check is whether your TCD can survive 30 minutes with a senior architect.